ŠTUDENT: ENEJ VOLMUT

1. DELO Z OBSTOJEČIM VIRTUALNIM RAČUNALNIKOM

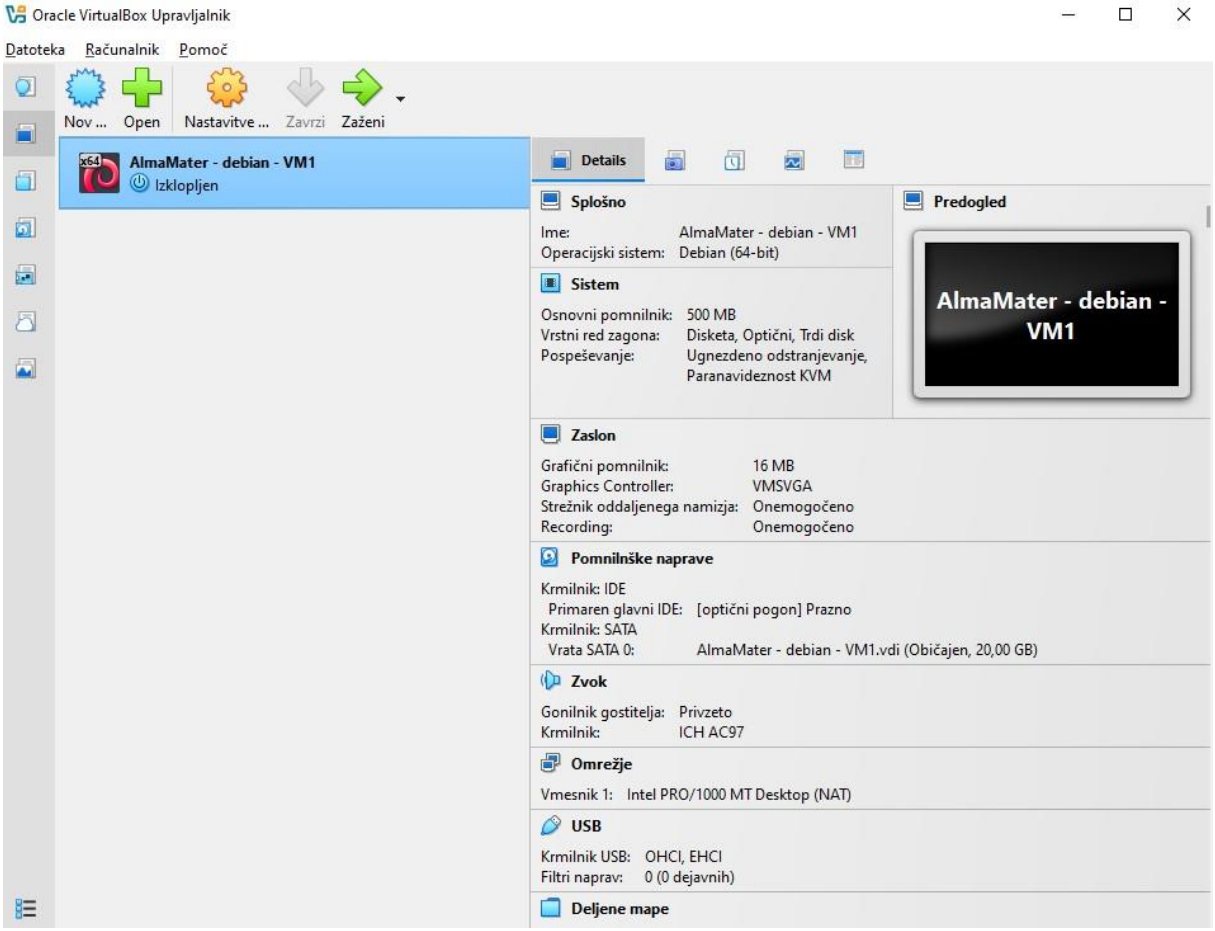
Opis postopka in uporabljeni ukazi:

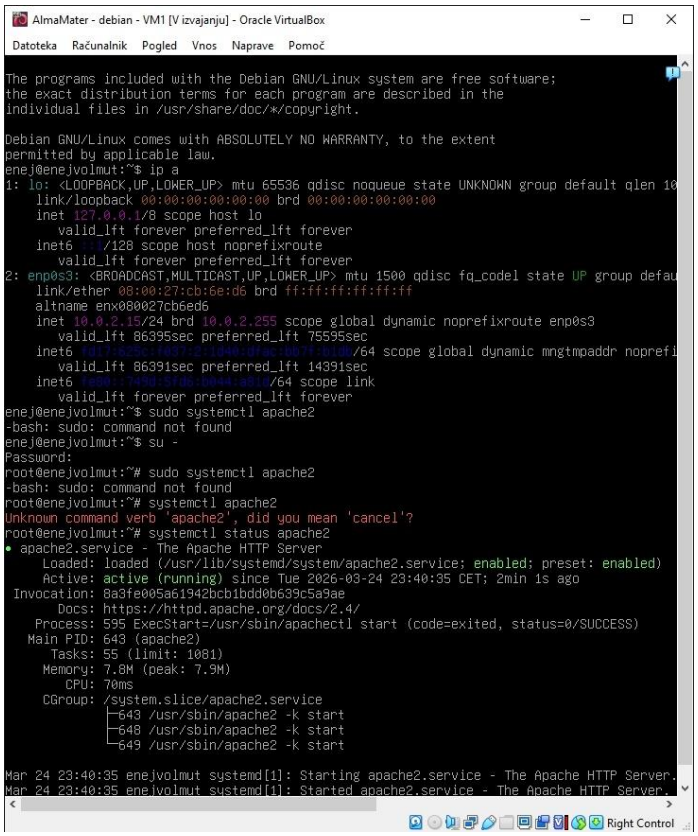
1. Najprej sem v nastavitvah virtualnega računalnika pod zavihkom Sistem nastavil količino delovnega pomnilnika na 500 MB. V zavihku Omrežje sem izbral način NAT in nato zagnal stroj.
2. V terminal sem se prijavil kot navadni uporabnik »enej«.
3. Preveril sem status spletnega strežnika z ukazom »systemctl status apache2«. Videl sem, da deluje.
4. Preveril sem IP naslov z ukazom »ip a«. Ugotovil sem, da v NAT načinu vmesnik enp0s3 nima dodeljenega IPv4 naslova.
5. Na fizičnem računalniku sem v ukaznem pozivu preveril IP naslov z ukazom »ipconfig«. Moj fizični IP je bil 192.168.0.15.
6. Poskusil sem dostopati do virtualnega stroja preko SSH in brskalnika, vendar povezava ni bila mogoča.
7. Virtualni stroj sem pravilno izklopil in v nastavitvah spremenil omrežje na Bridged (povezan vmesnik).
8. Po ponovnem zagonu sem ponovno preveril IP naslov (ip a). Debian je prejel naslov 192.168.0.27, ki je v istem omrežju kot fizični računalnik.
9. Uspešno sem dostopal do spletne strani preko brskalnika in se povezal preko SSH.

Opažanja in razlage

NAT način: Pri nastavitvi NAT je bil virtualni računalnik izoliran, ker vmesnik ni prejel IPv4 naslova, komunikacija sploh ni bila mogoča. Tudi če bi naprava prejela običajen NAT IP, dostop iz fizičnega računalnika ne bi deloval, saj NAT deluje kot enosmerna zaščita.

Bridged način: V Bridged načinu se je virtualni računalnik obnašal kot samostojna naprava v mojem domačem omrežju. Dobil je IP naslov od istega usmerjevalnika kot moj fizični računalnik. Ker sta bila oba v istem lokalnem omrežju, sta lahko neposredno komunicirala, zato sta SSH in brskalnik takoj delovala.


Oracle VirtualBox Upravljalnik
Datoteka Računalnik Pomoč
Nov ... Open Nastavitve ... Zavrzi Zaženi
AlmaMater - debian - VM1
Izklopljen
Details
Splošno
Ime: AlmaMater - debian - VM1
Operacijski sistem: Debian (64-bit)
Sistem
Osnovni pomnilnik: 500 MB
Vrstni red zagona: Disketa, Optični, Trdi disk
Pospeševanje: Ugnezdeno odstranjevanje, Paranavideznost KVM
Predogled
AlmaMater - debian - VM1
Zaslon
Grafični pomnilnik: 16 MB
Graphics Controller: VMSVGA
Strežnik oddaljenega namizja: Onemogočeno
Recording: Onemogočeno
Pomnilnške naprave
Krmilnik: IDE
Primaren glavni IDE: [optični pogon] Prazno
Krmilnik: SATA
Vrata SATA 0: AlmaMater - debian - VM1.vdi (Običajen, 20,00 GB)
Zvok
Gonilnik gostitelja: Privzeto
Krmilnik: ICH AC97
Omrežje
Vmesnik 1: Intel PRO/1000 MT Desktop (NAT)
USB
Krmilnik USB: OHCI, EHCI
Filtri naprav: 0 (0 dejavnih)
Deljene mape


AlmaMater - debian - VM1 [V izvajanju] - Oracle VirtualBox
Datoteka Računalnik Pogled Vnos Naprave Pomoč
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
enej@enejvolmut:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 10
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group defau
link/ether 08:00:27:cb:6e:d6 brd ff:ff:ff:ff:ff:ff
altname enx080027cb6ed6
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
valid_lft 86395sec preferred_lft 75595sec
inet6 [illegible]/64 scope global dynamic mngtmpaddr noprefi
valid_lft 86391sec preferred_lft 14391sec
inet6 [illegible]/64 scope link
valid_lft forever preferred_lft forever
enej@enejvolmut:~$ sudo systemctl apache2
-bash: sudo: command not found
enej@enejvolmut:~$ su -
Password:
root@enejvolmut:~# sudo systemctl apache2
-bash: sudo: command not found
root@enejvolmut:~# systemctl apache2
Unknown command verb 'apache2', did you mean 'cancel'?
root@enejvolmut:~# systemctl status apache2
● apache2.service - The Apache HTTP Server
Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
Active: active (running) since Tue 2026-03-24 23:40:35 CET; 2min 1s ago
Invocation: 8a3fe005a61942bcb1bdd0b639c5a9ae
Docs: https://httpd.apache.org/docs/2.4/
Process: 595 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
Main PID: 643 (apache2)
Tasks: 55 (limit: 1081)
Memory: 7.8M (peak: 7.9M)
CPU: 70ms
CGroup: /system.slice/apache2.service
643 /usr/sbin/apache2 -k start
648 /usr/sbin/apache2 -k start
649 /usr/sbin/apache2 -k start
Mar 24 23:40:35 enejvolmut systemd[1]: Starting apache2.service - The Apache HTTP Server.
Mar 24 23:40:35 enejvolmut systemd[1]: Started apache2.service - The Apache HTTP Server.
Right Control

Select Ukazni poziv

```
Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::21d5:3378:aba7:b829%16
   IPv4 Address. . . . . . . . . . . : 192.168.0.15
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.0.1

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::da05:49b7:a091:8ef4%8
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Wi-Fi:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Povezava z lokalnim omrežjem* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Povezava z lokalnim omrežjem* 10:

   Media State . . . . . . . . . . . : Media disconnected
```

PuTTY Configuration

Category:

- Session
  - Logging
- Terminal
  - Keyboard
  - Bell
  - Features
- Window
  - Appearance
  - Behaviour
  - Translation
  - Selection
  - Colours
- Connection
  - Data
  - Proxy
  - SSH
  - Serial
  - Telnet
  - Rlogin
  - SUPDUP

Basic options for your PuTTY session

Specify the destination you want to connect to

Host Name (or IP address): 192.168.0.15

Port: 22

Connection type: (•) SSH ( ) Serial ( ) Other: Telnet

Load, save or delete a stored session

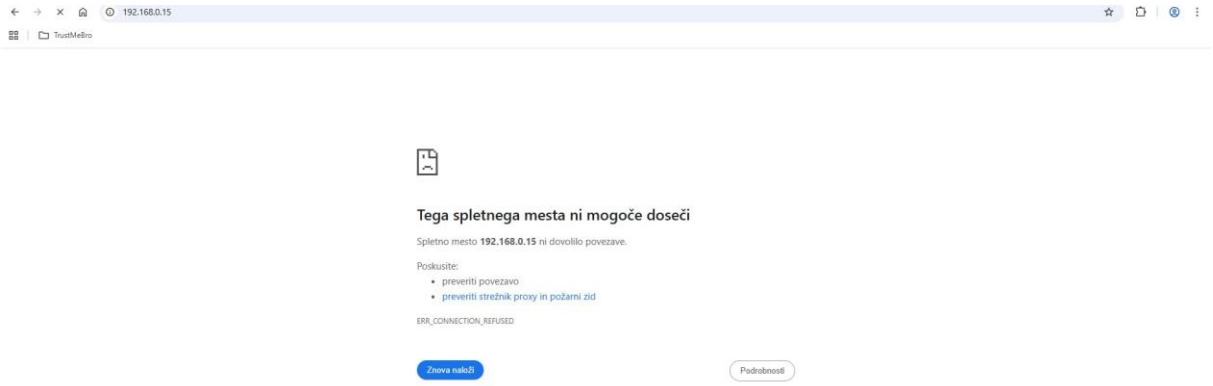
Saved Sessions

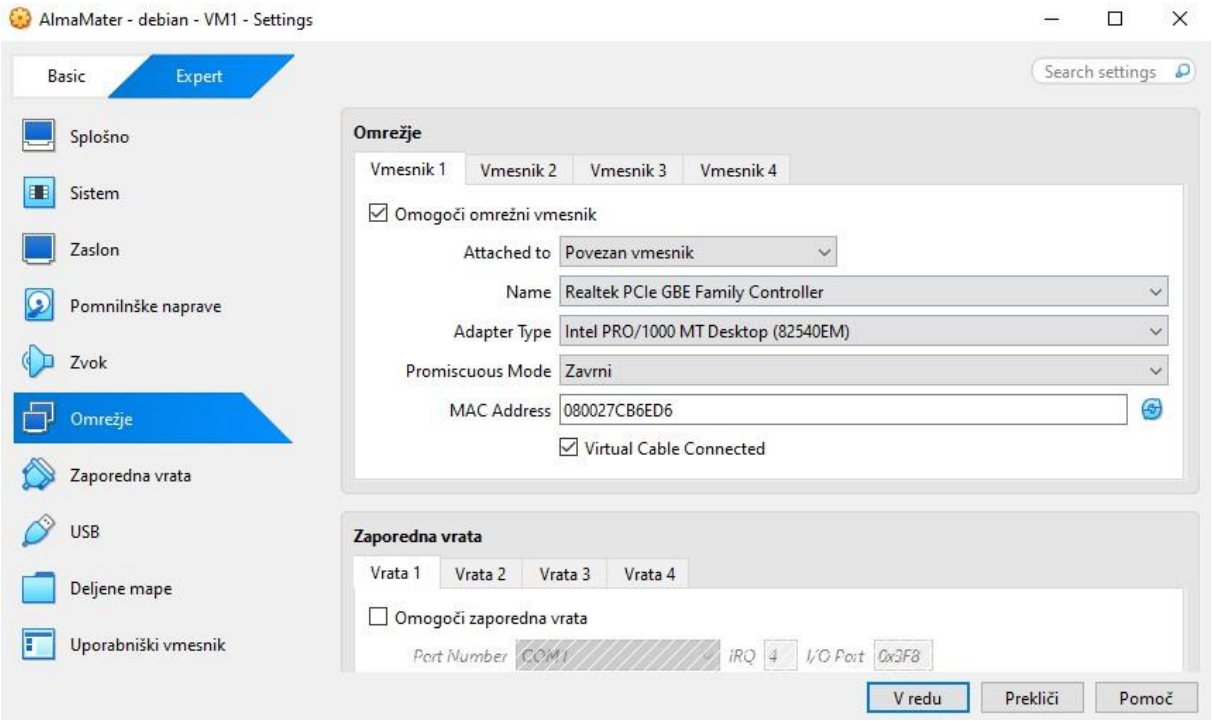
Default Settings
almaMaterDebian2026Lokalni
almaStudent

Load | Save | Delete

Close window on exit: ( ) Always ( ) Never (•) Only on clean exit

About | Help | Open | Cancel


192.168.0.15
Tega spletnega mesta ni mogoče doseči
Spletno mesto 192.168.0.15 ni dovolilo povezave.
Poskusite:
preveriti povezavo
preveriti strežnik proxy in požarni zid
ERR_CONNECTION_REFUSED
Znova naloži
Podrobnosti


AlmaMater - debian - VM1 - Settings
Basic
Expert
Search settings
Splošno
Sistem
Zaslon
Pomnilnške naprave
Zvok
Omrežje
Zaporedna vrata
USB
Deljene mape
Uporabniški vmesnik
Omrežje
Vmesnik 1
Vmesnik 2
Vmesnik 3
Vmesnik 4
Omogoči omrežni vmesnik
Attached to
Povezan vmesnik
Name
Realtek PCIe GBE Family Controller
Adapter Type
Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode
Zavrni
MAC Address
080027CB6ED6
Virtual Cable Connected
Zaporedna vrata
Vrata 1
Vrata 2
Vrata 3
Vrata 4
Omogoči zaporedna vrata
Port Number
COM1
IRQ
4
I/O Port
0x3F8
V redu
Prekliči
Pomoč

AlmaMater - debian - VM1 [V izvajanju] - Oracle VirtualBox

Datoteka Računalnik Pogled Vnos Naprave Pomoč

```
Debian GNU/Linux 13 enejvolmut tty1

enejvolmut login: enje
Password:
Login incorrect

enejvolmut login: enej
Password:
Linux enejvolmut 6.12.74+deb13+1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
enej@enejvolmut:~$ su -
Password:
root@enejvolmut:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:cb:6e:d6 brd ff:ff:ff:ff:ff:ff
    altname enx080027cb6ed6
    inet 192.168.0.27/24 brd 192.168.0.255 scope global dynamic noprefixroute enp0s3
       valid_lft 3528sec preferred_lft 3078sec
    inet6 fe80::f49d:5fd8:b4e4:a8d7/64 scope link
       valid_lft forever preferred_lft forever
root@enejvolmut:~#
```

Mouse integration ...

Don't show again

Auto capture keyboard ...

Don't show again

Right Control

Ni varno 192.168.0.27

TrustMeBro

Spletna stran za predmet Oblacne storitve

Avtor: Enej Volmut

2. NOV VIRTUALNI STREŽNIK (LOKALNO)

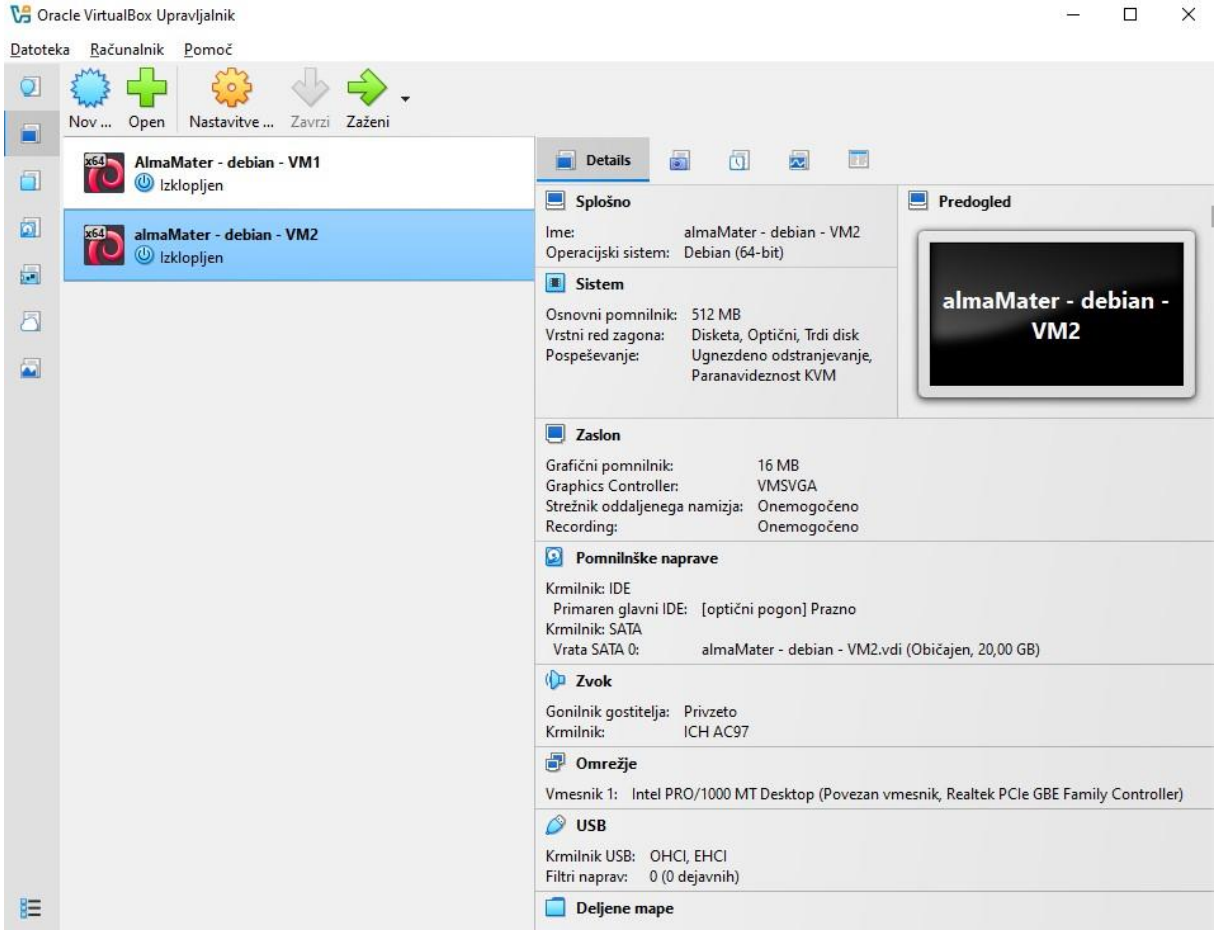
Opis postopka in uporaba ukazov:

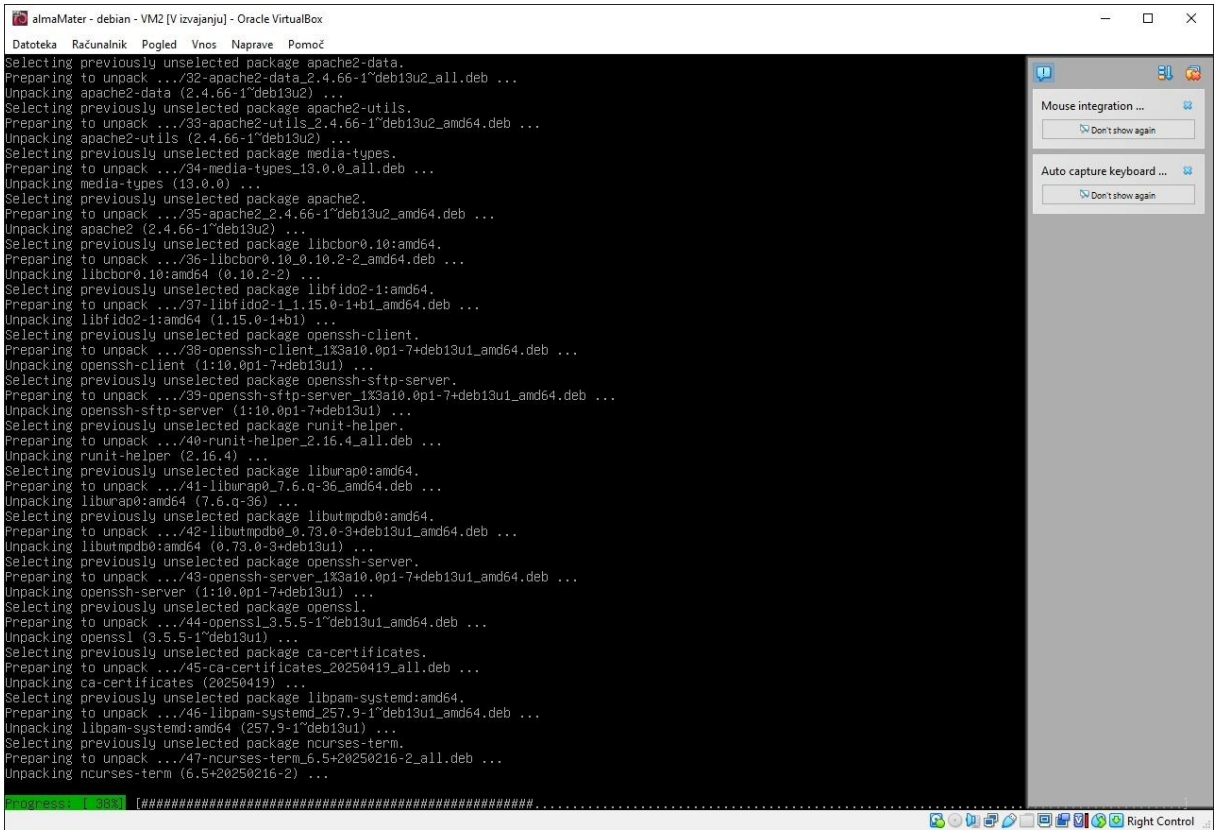
1. Sprva sem v VirtualBoxu ustvaril nov virtualni stroj (VM2). Dodelil sem mu 1 CPU in 512 MB RAM. Omrežni vmesnik sem nastavil na Bridged Adapter, kar omogoča, da virtualni stroj prejme IP naslov neposredno iz mojega lokalnega omrežja.
2. Namestil sem operacijski sistem Debian. Med namestitvijo sem ustvaril uporabnika enej in nastavil geslo za skrbnika (root).
3. Prijavil sem se kot root z ukazom »su -«. Posodobil sem sezname paketov in namestil zahtevani storitvi apt update in apt install ssh apache2 -y
4. Preverjanje delovanja: Z ukazom »ip a« sem preveril omrežni naslov in ugotovil, da ima strežnik IP 192.168.0.38. S pomočjo »systemctl status ssh« in »systemctl status apache2« sem potrdil, da sta obe storitvi v aktivnem stanju.
5. Iz Windows PowerShell sem uspešno povezal z ukazom »ssh enej@192.168.0.38«. Ob vpisu IP naslova v brskalnik se je prikazala privzeta stran Apache2 Debian Default Page. To je tehnična stran z napisom, ki potrjuje, da strežnik deluje, vendar v mapi še ni moje vsebine.
6. Vklopil sem oba virtualna računalnika. Na prvem računalniku (VM1) sem uporabil ukaz scp za prenos datotek na novi strežnik: »scp -r /home/enej/* »enej@192.168.0.38:/home/enej/«
7. Na drugem strežniku sem nato datoteke premaknil v sistemsko mapo: »mv /home/enej/* /var/www/html/«
8. V brskalniku se je vsebina spremenila. Namesto privzete Apache strani se zdaj na naslovu 192.168.0.38 prikazuje moja osebna spletna stran, kar dokazuje, da so na novem strežniku uspešno preneseni podatki.

Opažanja in razlage

Pri delu sem opazil, da Debian privzeto ne vključi navadnega uporabnika v skupino sudo, zato sem moral za vse sistemske spremembe uporabljati ukaz su - in geslo za root.

Uporaba Bridged omrežja je bila pomembna, saj sta oba virtualna stroja in fizični računalnik del istega podomrežja, kar je omogočilo prenos datotek s protokolom scp. Pri prenosu sem moral biti pazljiv na poti, saj sem sprva pomotoma poskušal premikati datoteke na napačnem računalniku, kar sem rešil s preverjanjem vsebine z ukazom ls.


Oracle VirtualBox Upravljalnik
Datoteka Računalnik Pomoč
Nov ... Open Nastavitve ... Zavrzi Zaženi
AlmaMater - debian - VM1
Izklopljen
almaMater - debian - VM2
Izklopljen
Details
Splošno
Ime: almaMater - debian - VM2
Operacijski sistem: Debian (64-bit)
Sistem
Osnovni pomnilnik: 512 MB
Vrstni red zagona: Disketa, Optični, Trdi disk
Pospeševanje: Ugnezdeno odstranjevanje, Paranavideznost KVM
Predogled
almaMater - debian - VM2
Zaslon
Grafični pomnilnik: 16 MB
Graphics Controller: VMSVGA
Strežnik oddaljenega namizja: Onemogočeno
Recording: Onemogočeno
Pomnilnške naprave
Krmilnik: IDE
Primaren glavni IDE: [optični pogon] Prazno
Krmilnik: SATA
Vrata SATA 0: almaMater - debian - VM2.vdi (Običajen, 20,00 GB)
Zvok
Gonilnik gostitelja: Privzeto
Krmilnik: ICH AC97
Omrežje
Vmesnik 1: Intel PRO/1000 MT Desktop (Povezan vmesnik, Realtek PCIe GBE Family Controller)
USB
Krmilnik USB: OHCI, EHCI
Filtri naprav: 0 (0 dejavnih)
Deljene mape


almaMater - debian - VM2 [V izvajanju] - Oracle VirtualBox
Datoteka Računalnik Pogled Vnos Naprave Pomoč
Selecting previously unselected package apache2-data.
Preparing to unpack .../32-apache2-data_2.4.66-1~deb13u2_all.deb ...
Unpacking apache2-data (2.4.66-1~deb13u2) ...
Selecting previously unselected package apache2-utils.
Preparing to unpack .../33-apache2-utils_2.4.66-1~deb13u2_amd64.deb ...
Unpacking apache2-utils (2.4.66-1~deb13u2) ...
Selecting previously unselected package media-types.
Preparing to unpack .../34-media-types_13.0.0_all.deb ...
Unpacking media-types (13.0.0) ...
Selecting previously unselected package apache2.
Preparing to unpack .../35-apache2_2.4.66-1~deb13u2_amd64.deb ...
Unpacking apache2 (2.4.66-1~deb13u2) ...
Selecting previously unselected package libcbor0.10:amd64.
Preparing to unpack .../36-libcbor0.10_0.10.2-2_amd64.deb ...
Unpacking libcbor0.10:amd64 (0.10.2-2) ...
Selecting previously unselected package libfido2-1:amd64.
Preparing to unpack .../37-libfido2-1_1.15.0-1+b1_amd64.deb ...
Unpacking libfido2-1:amd64 (1.15.0-1+b1) ...
Selecting previously unselected package openssh-client.
Preparing to unpack .../38-openssh-client_1%3a10.0p1-7+deb13u1_amd64.deb ...
Unpacking openssh-client (1:10.0p1-7+deb13u1) ...
Selecting previously unselected package openssh-sftp-server.
Preparing to unpack .../39-openssh-sftp-server_1%3a10.0p1-7+deb13u1_amd64.deb ...
Unpacking openssh-sftp-server (1:10.0p1-7+deb13u1) ...
Selecting previously unselected package runit-helper.
Preparing to unpack .../40-runit-helper_2.16.4_all.deb ...
Unpacking runit-helper (2.16.4) ...
Selecting previously unselected package libwrap0:amd64.
Preparing to unpack .../41-libwrap0_7.6.q-36_amd64.deb ...
Unpacking libwrap0:amd64 (7.6.q-36) ...
Selecting previously unselected package libutmpdb0:amd64.
Preparing to unpack .../42-libutmpdb0_0.73.0-3+deb13u1_amd64.deb ...
Unpacking libutmpdb0:amd64 (0.73.0-3+deb13u1) ...
Selecting previously unselected package openssh-server.
Preparing to unpack .../43-openssh-server_1%3a10.0p1-7+deb13u1_amd64.deb ...
Unpacking openssh-server (1:10.0p1-7+deb13u1) ...
Selecting previously unselected package openssl.
Preparing to unpack .../44-openssl_3.5.5-1~deb13u1_amd64.deb ...
Unpacking openssl (3.5.5-1~deb13u1) ...
Selecting previously unselected package ca-certificates.
Preparing to unpack .../45-ca-certificates_20250419_all.deb ...
Unpacking ca-certificates (20250419) ...
Selecting previously unselected package libpam-systemd:amd64.
Preparing to unpack .../46-libpam-systemd_257.9-1~deb13u1_amd64.deb ...
Unpacking libpam-systemd:amd64 (257.9-1~deb13u1) ...
Selecting previously unselected package ncurses-term.
Preparing to unpack .../47-ncurses-term_6.5+20250216-2_all.deb ...
Unpacking ncurses-term (6.5+20250216-2) ...
Progress: [38%] [###.....
Mouse integration ...
Don't show again
Auto capture keyboard ...
Don't show again
Right Control

almaMater - debian - VM2 [V izvajanju] - Oracle VirtualBox

Datoteka Računalnik Pogled Vnos Naprave Pomoč

```
Creating user 'sshd' (sshd user) with UID 990 and GID 65534.
Created symlink '/etc/systemd/system/sshd.service' -> '/usr/lib/systemd/system/ssh.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' -> '/usr/lib/systemd/system/ssh.service'.
ssh.socket is a disabled or a static unit, not starting it.
Created symlink '/etc/systemd/system/ssh.service.wants/sshd-keygen.service' -> '/usr/lib/systemd/system/sshd-keygen.service'.
Created symlink '/etc/systemd/system/sshd.service.wants/sshd-keygen.service' -> '/usr/lib/systemd/system/sshd-keygen.service'.
Created symlink '/etc/systemd/system/sshd@.service.wants/sshd-keygen.service' -> '/usr/lib/systemd/system/sshd-keygen.service'.
Created symlink '/etc/systemd/system/ssh.socket.wants/sshd-keygen.service' -> '/usr/lib/systemd/system/sshd-keygen.service'.
Setting up dbus-user-session (1.16.2-2) ...
Setting up librtmp1:amd64 (2.4+20151223.gitfa8646d.1-2+b5) ...
Setting up ssh (1:10.0p1-7+deb13u1) ...
Setting up libcurl4t64:amd64 (8.14.1-2+deb13u2) ...
Setting up apache2-bin (2.4.66-1~deb13u2) ...
Setting up apache2 (2.4.66-1~deb13u2) ...
Enabling module mpm_event.
Enabling module authz_core.
Enabling module authz_host.
Enabling module authn_core.
Enabling module auth_basic.
Enabling module access_compat.
Enabling module authn_file.
Enabling module authz_user.
Enabling module alias.
Enabling module dir.
Enabling module autoindex.
Enabling module env.
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink '/etc/systemd/system/multi-user.target.wants/apache2.service' -> '/usr/lib/systemd/system/apache2.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service' -> '/usr/lib/systemd/system/apache-htcacheclean.service'.
Processing triggers for libc-bin (2.41-12+deb13u2) ...
Processing triggers for ca-certificates (20250419) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
root@enejvolmut:/home/enej#
```

Mouse integration ... Don't show again

Auto capture keyboard ... Don't show again

Right Control

almaMater - debian - VM2 [V izvajanju] - Oracle VirtualBox

Datoteka Računalnik Pogled Vnos Naprave Pomoč

```
Setting up apache2-bin (2.4.66-1~deb13u2) ...
Setting up apache2 (2.4.66-1~deb13u2) ...
Enabling module mpm_event.
Enabling module authz_core.
Enabling module authz_host.
Enabling module authn_core.
Enabling module auth_basic.
Enabling module access_compat.
Enabling module authn_file.
Enabling module authz_user.
Enabling module alias.
Enabling module dir.
Enabling module autoindex.
Enabling module env.
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink '/etc/systemd/system/multi-user.target.wants/apache2.service' -> '/usr/lib/systemd/system/apache2.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service' -> '/usr/lib/systemd/system/apache-htcacheclean.service'.
Processing triggers for libc-bin (2.41-12+deb13u2) ...
Processing triggers for ca-certificates (20250419) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
root@enejvolmut:/home/enej# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:55:93:ed brd ff:ff:ff:ff:ff:ff
    altname enx0800275593ed
    inet 192.168.0.38/24 brd 192.168.0.255 scope global dynamic noprefixroute enp0s3
       valid_lft 3119sec preferred_lft 2669sec
    inet6 [illegible]/64 scope link
       valid_lft forever preferred_lft forever
root@enejvolmut:/home/enej#
```

Mouse integration ... Don't show again

Auto capture keyboard ... Don't show again

Right Control

almaMater - debian - VM2 [V izvajanju] - Oracle VirtualBox

Datoteka Računalnik Pogled Vnos Naprave Pomoč

```
root@enejvolmut:/home/enej# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:55:93:ed brd ff:ff:ff:ff:ff:ff
    altname enx0800275593ed
    inet 192.168.0.38/24 brd 192.168.0.255 scope global dynamic noprefixroute enp0s3
       valid_lft 3119sec preferred_lft 2669sec
    inet6 fe80::d5c9:ee5b:390e:5614/64 scope link
       valid_lft forever preferred_lft forever
root@enejvolmut:/home/enej# systemctl status ssh
* ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-03-24 19:42:17 CET; 3min 10s ago
 Invocation: a62e913a617e4f919e3770f7a3a06f22
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 3644 (sshd)
      Tasks: 1 (limit: 506)
     Memory: 1.3M (peak: 1.8M)
        CPU: 31ms
     CGroup: /system.slice/ssh.service
             `-3644 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 24 19:42:17 enejvolmut systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Mar 24 19:42:17 enejvolmut sshd[3644]: Server listening on 0.0.0.0 port 22.
Mar 24 19:42:17 enejvolmut sshd[3644]: Server listening on :: port 22.
Mar 24 19:42:17 enejvolmut systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
root@enejvolmut:/home/enej# systemctl status apache2
* apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-03-24 19:42:21 CET; 3min 30s ago
 Invocation: a374cd7be21b4c96a7d2fb52e93d1554
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 3987 (apache2)
      Tasks: 55 (limit: 506)
     Memory: 4.9M (peak: 5.1M)
        CPU: 51ms
     CGroup: /system.slice/apache2.service
             |-3987 /usr/sbin/apache2 -k start
             |-3988 /usr/sbin/apache2 -k start
             `-3992 /usr/sbin/apache2 -k start

Mar 24 19:42:21 enejvolmut systemd[1]: Starting apache2.service - The Apache HTTP Server...
Mar 24 19:42:21 enejvolmut systemd[1]: Started apache2.service - The Apache HTTP Server.
root@enejvolmut:/home/enej# _
```

Right Control

Ni varno 192.168.0.38

TrustMeBro

Apache2 Debian Default Page

debian

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Debian systems. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Debian's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Debian tools. The configuration system is **fully documented in /usr/share/doc/apache2/README.Debian.gz**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Debian systems is as follows:

```
/etc/apache2/
|-- apache2.conf
|       `--  ports.conf
|-- mods-enabled
|       |-- *.load
|       `-- *.conf
|-- conf-enabled
|       `-- *.conf
|-- sites-enabled
|       `-- *.conf
```

- `apache2.conf` is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.
- `ports.conf` is always included from the main configuration file. It is used to determine the listening ports for incoming connections, and this file can be customized anytime.
- Configuration files in the `mods-enabled/`, `conf-enabled/` and `sites-enabled/` directories contain particular configuration snippets which manage modules, global configuration fragments, or virtual host configurations, respectively.
- They are activated by symlinking available configuration files from their respective *-available/ counterparts. These should be managed by using our helpers `a2enmod`, `a2dismod`, `a2ensite`, `a2dissite`, and `a2enconf`, `a2disconf` . See their respective man pages for detailed information.
- The binary is called apache2. Due to the use of environment variables, in the default configuration, apache2 needs to be started/stopped with `/etc/init.d/apache2` or `apache2ctl`.

enej@enejvolmut: ~

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\Asus>  ssh enej@192.168.0.38
enej@192.168.0.38's password:
Linux enejvolmut 6.12.74+deb13+1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
enej@enejvolmut: $
```

AlmaMater - debian - VM1 [V izvajanju] - Oracle VirtualBox

Datoteka Računalnik Pogled Vnos Naprave Pomoč

```
Debian GNU/Linux 13 enejvolmut tty1

enejvolmut login: enej
Password:
Linux enejvolmut 6.12.74+deb13+1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86_6

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
enej@enejvolmut:~$ su -
Password:
root@enejvolmut:~# scp -r /var/www/html/* enej@192.168.0.38:/home/enej
The authenticity of host '192.168.0.38 (192.168.0.38)' can't be established.
ED25519 key fingerprint is SHA256:XpuHvbldTpL8+8PzwzUGpzGvPWvbXruS8GVe7yM3Uvw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.0.38' (ED25519) to the list of known hosts.
Connection closed by 192.168.0.38 port 22
scp: Connection closed
root@enejvolmut:~#
```

Right Control

AlmaMater - debian - VM1 [V izvajanju] - Oracle VirtualBox

Datoteka Računalnik Pogled Vnos Naprave Pomoč

```
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.0.38' (ED25519) to the list of known hosts.
Connection closed by 192.168.0.38 port 22
scp: Connection closed
root@enejvolmut:~# pwd
/root
root@enejvolmut:~# ls -R /home/enej
/home/enej:
splet

/home/enej/splet:
DSC00574.JPG  index.html
root@enejvolmut:~# mv /home/enej/* /var/www/html/
root@enejvolmut:~# ls /var/www/html
enej  index.html  slika.jpg  splet
root@enejvolmut:~#
root@enejvolmut:~# mv /var/www/html/* /home/enej/
root@enejvolmut:~# chown -R enej:enej /home/enej/
root@enejvolmut:~# scp -r /var/www/html/* enej@192.168.0.38:/home/enej/
scp: stat local "/var/www/html/*": No such file or directory
root@enejvolmut:~# pwd
/root
root@enejvolmut:~# ls /var/www/html
root@enejvolmut:~# ls /var/www/html/
root@enejvolmut:~# ls /home/enej/
enej  index.html  slika.jpg  splet
root@enejvolmut:~# scp -r /home/enej/* enej@192.168.0.38:/home/enej/
enej@192.168.0.38's password:
.bashrc
.profile
.bash_history
index.html
DSC00574.JPG
.bash_logout
index.html
slika.jpg
index.html
DSC00574.JPG
root@enejvolmut:~# _
```

Right Control

almaMater - debian - VM2 [V izvajanju] - Oracle VirtualBox

Datoteka Računalnik Pogled Vnos Naprave Pomoč

```
Debian GNU/Linux 13 enejvolmut tty1

enejvolmut login: enej
Password:
Linux enejvolmut 6.12.74+deb13+1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
enej@enejvolmut:~$ su -
Password:
root@enejvolmut:~# mv /home/enej/* /var/www/html
mv: cannot stat '/home/enej/*': No such file or directory
root@enejvolmut:~# ls /home/enej/
root@enejvolmut:~# pwd
/root
root@enejvolmut:~# ls /home/enej/
enej  index.html  slika.jpg  splet
root@enejvolmut:~# _
```

Right Control

almaMater - debian - VM2 [V izvajanju] - Oracle VirtualBox

Datoteka Računalnik Pogled Vnos Naprave Pomoč

```
Debian GNU/Linux 13 enejvolmut tty1

enejvolmut login: enej
Password:
Linux enejvolmut 6.12.74+deb13+1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
enej@enejvolmut:~$ su -
Password:
root@enejvolmut:~# mv /home/enej/* /var/www/html
mv: cannot stat '/home/enej/*': No such file or directory
root@enejvolmut:~# ls /home/enej/
root@enejvolmut:~# pwd
/root
root@enejvolmut:~# ls /home/enej/
enej  index.html  slika.jpg  splet
root@enejvolmut:~# mv /home/enej/index.html /var/www/html
root@enejvolmut:~# mv /home/enej/slike /var/www/html
mv: cannot stat '/home/enej/slike': No such file or directory
root@enejvolmut:~# mv /home/enej/slika.jpg /var/www/html
root@enejvolmut:~# chown -R www-data:www-data /var/www/html
root@enejvolmut:~# chown -R 755 /var/www/html/
root@enejvolmut:~#
```

Right Control

Ni varno 192.168.0.38

Spletna stran za predmet Oblacne storitve

Avtor: Enej Volmut

3. VIRTUALNI STREŽNIK V OBLAKU (AWS EC2)

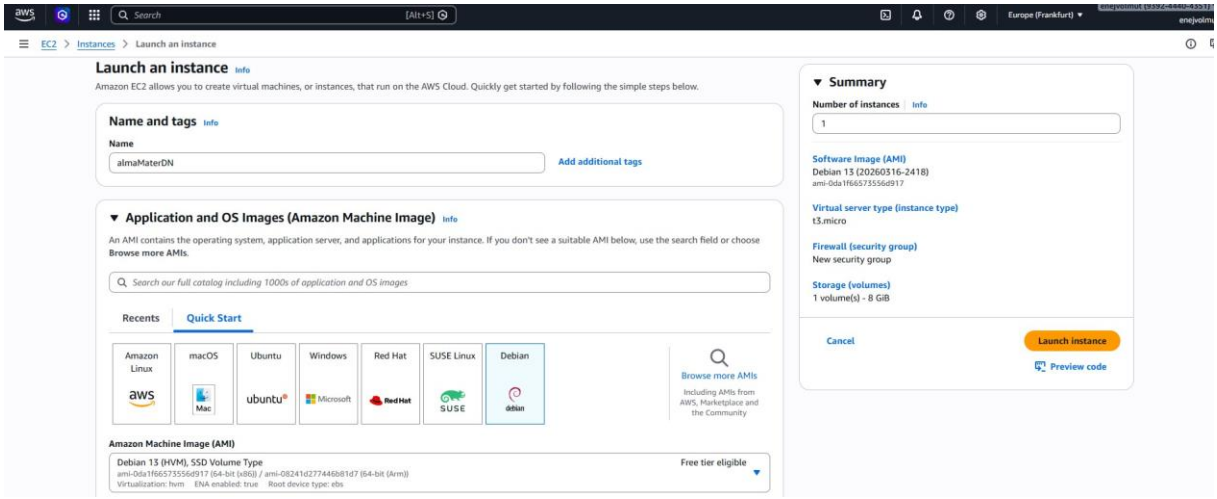
Opis postopka in uporaba ukazov:

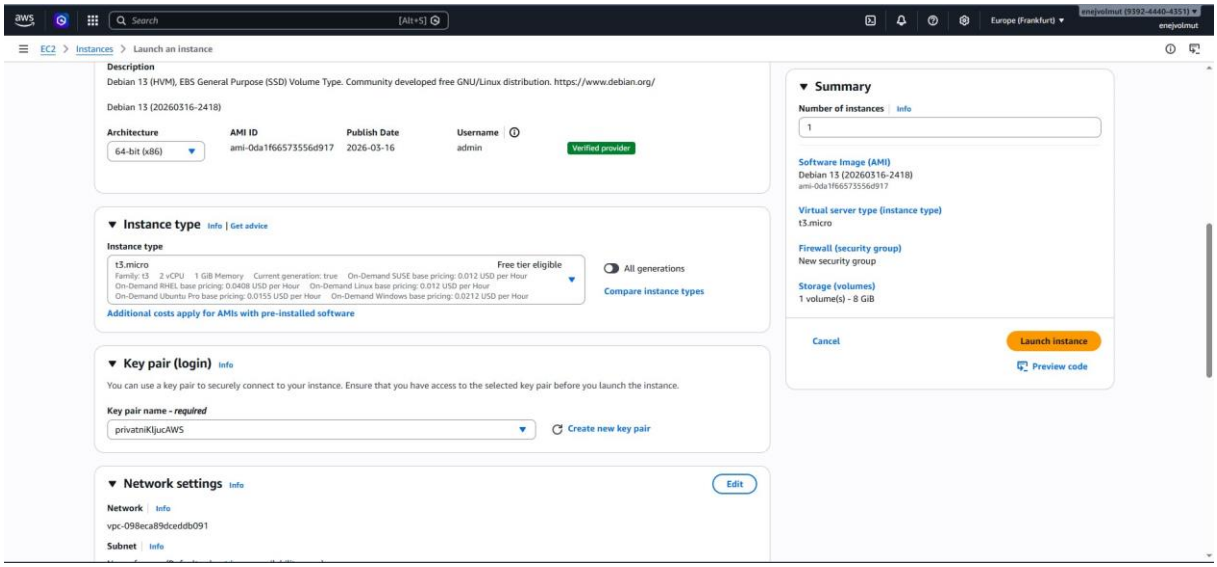
1. V nadzorni plošči AWS sem preko možnosti "Launch Instance" ustvaril nov virtualni strežnik z imenom almaMaterDN, kjer sem za operacijski sistem izbral Debian, uporabil privzeto strojno opremo t3.micro ter za varno avtentikacijo določil obstoječi par ključev privatniKljucAWS.
2. S pomočjo odjemalca PuTTY sem vzpostavil oddaljeno povezavo, tako da sem vpisal javni IP naslov in v nastavitvah avtentikacije uvozil ustrezno .ppk datoteko ključa, kar mi je omogočilo prijavo v terminal kot uporabnik admin.
3. Po uspešni prijavi sem z ukazi »sudo systemctl status ssh« in »sudo systemctl status apache2« in »ip a« preveril stanje sistema. Ugotovil sem, da SSH servis deluje, spletni strežnik Apache pa še ni nameščen, ukaz za omrežje pa je izpisal IP naslov.
4. Ob vpisu javnega IP naslova in brskalnika fizičnega računalnika se stran pričakovano ni naložila, saj na strežniku še ni bilo aktivne spletne storitve, hkrati pa AWS požarni zid še ni bil nastavljen za prepustnost spletnega prometa.
5. V terminalu instance sem z ukazoma »sudo apt update« in »sudo apt install apache2 -y« namestil spletni strežnik, kjer je viden proces prenašanja in konfiguracije vseh potrebnih programskih paketov za delovanje HTTP protokola.
6. Po namestitvi spletnega strežnika sem ponovno osvežil brskalnik, vendar se stanje ni spremenilo in dostop je bil še vedno onemogočen, ker so bila vrata (port 80) na nivoju AWS varnostne skupine še vedno zaprta, kar dokazuje slika.
7. V AWS konzoli sem pod zavihkom »Security« uredil vstopna pravila pod »Inbound rules« za svojo instanco in dodal novo pravilo za HTTP na portu 80, pri čemer sem za vir določil 0.0.0.0/0, s čimer sem dovolil spletni promet.
8. Takoj po shranitvi varnostnih pravil se je ob osvežitvi brskalnika naložila privzeta stran »Apache2 Debian Default Page«, kar je pomenilo, da spletni strežnik deluje.
9. Ustvaril sem drugo instanco ter nanjo namestil Apache, nato pa iz lokalnega PowerShella z ukazom »scp –i index.html admin@18.153.100.6:/home/admin/« privatni ključ prenesel na svojo spletno stran, kjer sem moral prej z ukazom icacls popraviti varnostne pravice ključa na Windowsu.
10. Po prenosu sem datoteke na drugi instanci z ukazom »sudo mv index.html /var/www/html/« premaknil v mapo strežnika, kjer se je uspešno prikazala moja spletna stran na novem IP naslovu.

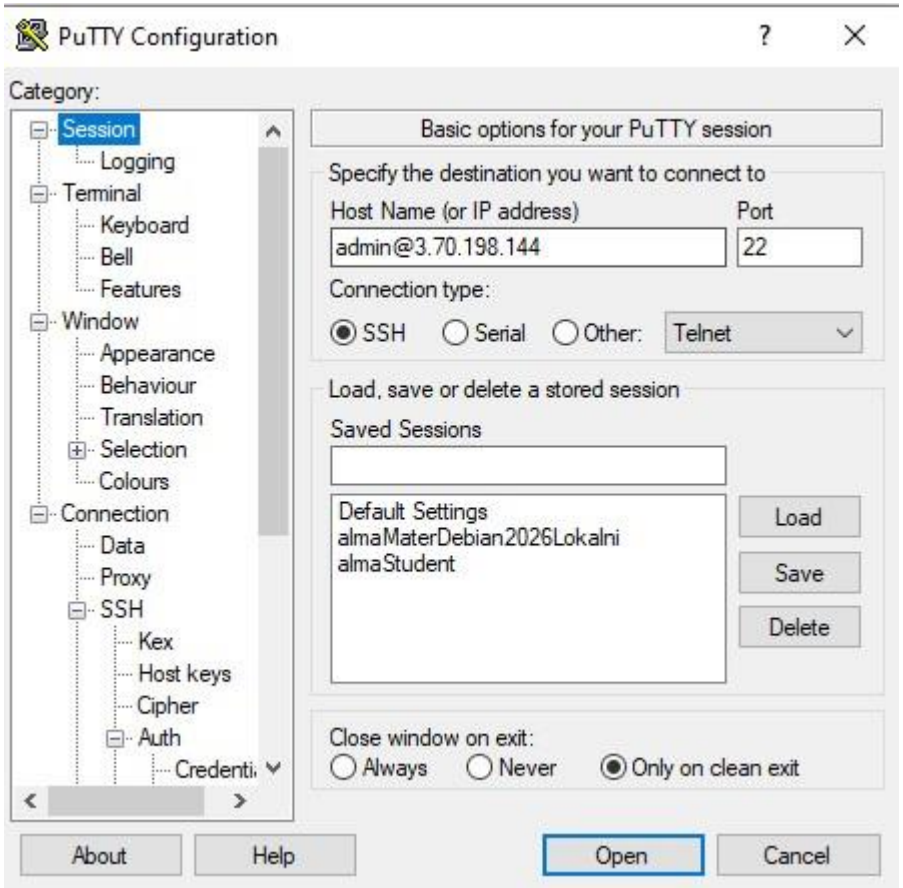
Opažanja in razlage

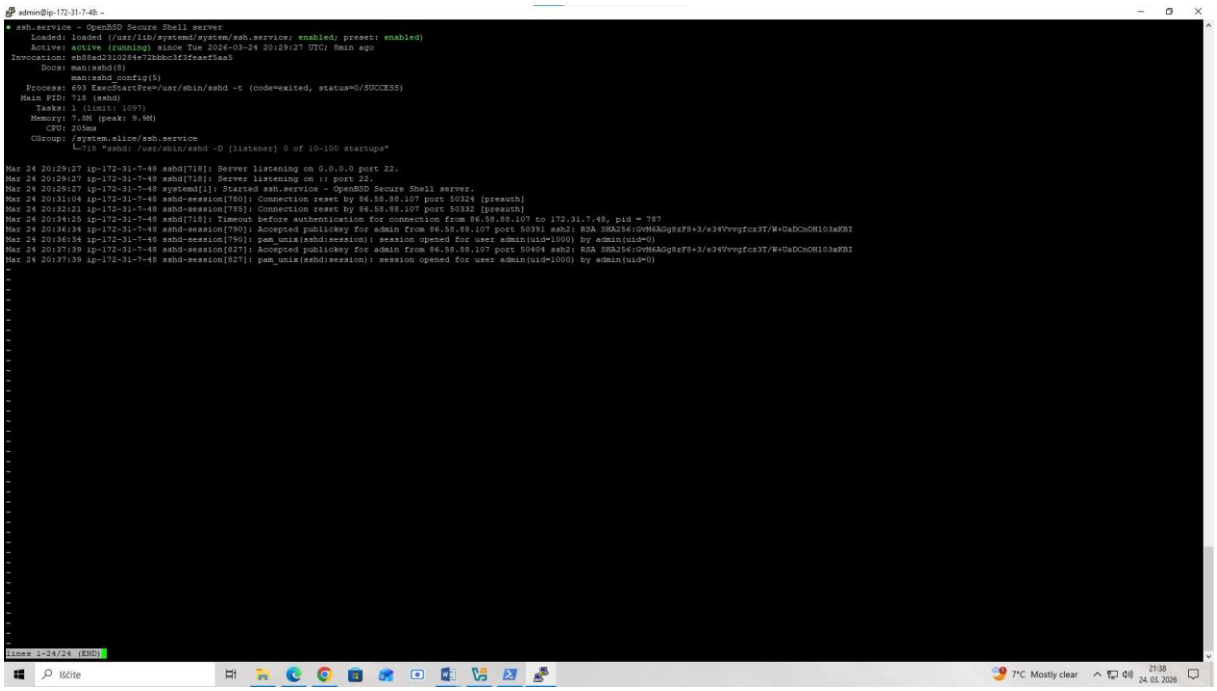
Ob prvem poskusu dostopa do strežnika sem opazil, da brskalnik javi napako "Timed out", kar je posledica, da na sveži instanci Debian še ni bil nameščen spletni strežnik. Tudi po namestitvi Apacheja dostop ni bil mogoč, dokler nisem v AWS konzoli ročno dodal pravila za HTTP promet, kar pomeni, da oblačna infrastruktura deluje kot zunanji požarni zid, ki je neodvisen od nastavitev znotraj samega operacijskega sistema.

Pri prenosu datotek z ukazom »scp« sem ugotovil, da SSH protokol zahteva strogo omejene pravice nad privatnim ključem, zaradi česar sem moral z ukazom »icacls« ročno preprečil dostop drugim uporabnikom sistema Windows. Moja spletna stran se je pravilno prikazala šele po premiku datotek iz domače mape v »/var/www/html/«, kar pomeni, da je nujna pravilna struktura map in usklajenost sistemskih pravic na strežniku.






PuTTY Configuration
Category:
Session
Logging
Terminal
Keyboard
Bell
Features
Window
Appearance
Behaviour
Translation
Selection
Colours
Connection
Data
Proxy
SSH
Kex
Host keys
Cipher
Auth
Credenti
Basic options for your PuTTY session
Specify the destination you want to connect to
Host Name (or IP address)
Port
admin@3.70.198.144
22
Connection type:
SSH
Serial
Other:
Telnet
Load, save or delete a stored session
Saved Sessions
Default Settings
almaMaterDebian2026Lokalni
almaStudent
Load
Save
Delete
Close window on exit:
Always
Never
Only on clean exit
About
Help
Open
Cancel


admin@ip-172-31-7-48: ~
● ssh.service - OpenBSD Secure Shell server
Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
Active: active (running) since Tue 2026-03-24 20:29:27 UTC; 8min ago
Invocation: eb88ad2210264e72bbbc3f3feaefbaa5
Docs: man:sshd(8)
man:sshd_config(5)
Process: 693 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
Main PID: 718 (sshd)
Tasks: 1 (limit: 1097)
Memory: 7.8M (peak: 9.8M)
CPU: 205ms
CGroup: /system.slice/ssh.service
└718 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
Mar 24 20:29:27 ip-172-31-7-48 sshd[718]: Server listening on 0.0.0.0 port 22.
Mar 24 20:29:27 ip-172-31-7-48 sshd[718]: Server listening on :: port 22.
Mar 24 20:29:27 ip-172-31-7-48 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Mar 24 20:31:04 ip-172-31-7-48 sshd-session[785]: Connection reset by 86.58.88.107 port 50324 [preauth]
Mar 24 20:32:21 ip-172-31-7-48 sshd-session[785]: Connection reset by 86.58.88.107 port 50332 [preauth]
Mar 24 20:34:25 ip-172-31-7-48 sshd[718]: Timeout before authentication for connection from 86.58.88.107 to 172.31.7.48, pid = 787
Mar 24 20:36:34 ip-172-31-7-48 sshd-session[790]: Accepted publickey for admin from 86.58.88.107 port 50391 ssh2: RSA SHA256:GvM6AGg8zF8+3/e34Vvvgfcz3T/W+UaDCnOH103xKBI
Mar 24 20:36:34 ip-172-31-7-48 sshd-session[790]: pam_unix(sshd:session): session opened for user admin(uid=1000) by admin(uid=0)
Mar 24 20:37:39 ip-172-31-7-48 sshd-session[827]: Accepted publickey for admin from 86.58.88.107 port 50404 ssh2: RSA SHA256:GvM6AGg8zF8+3/e34Vvvgfcz3T/W+UaDCnOH103xKBI
Mar 24 20:37:39 ip-172-31-7-48 sshd-session[827]: pam_unix(sshd:session): session opened for user admin(uid=1000) by admin(uid=0)
lines 1-24/24 (END)

admin@ip-172-31-7-48: ~

```
admin@ip-172-31-7-48:~$ sudo systemctl status apache2
Unit apache2.service could not be found.
admin@ip-172-31-7-48:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group defa
ult qlen 1000
    link/ether 0a:a4:51:eb:7f:8d brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname enx0aa451eb7f8d
    inet 172.31.7.48/20 metric 100 brd 172.31.15.255 scope global dynamic ens5
       valid_lft 2865sec preferred_lft 2865sec
    inet6 fe80::8a4:51ff:feeb:7f8d/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
admin@ip-172-31-7-48:~$
```

Ni varno 3.70.198.144

TrustMeBro

Tega spletnega mesta ni mogoče doseči

Spletno mesto **3.70.198.144** se ni odzvalo v ustreznem času.

Poskusite:

- preveriti povezavo
- preveriti strežnik proxy in požarni zid
- Izvajanje orodja Omrežna diagnostika Windows

ERR_CONNECTION_TIMED_OUT

Znova naloži

Podrobnosti

admin@ip-172-31-7-48: ~

```
      valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
      valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group defa
ult qlen 1000
    link/ether 0a:a4:51:eb:7f:8d brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname enx0aa451eb7f8d
    inet 172.31.7.48/20 metric 100 brd 172.31.15.255 scope global dynamic ens5
      valid_lft 2865sec preferred_lft 2865sec
    inet6 fe80::8a4:51ff:feeb:7f8d/64 scope link proto kernel_ll
      valid_lft forever preferred_lft forever
admin@ip-172-31-7-48:~$ sudo apt update
Get:1 file:/etc/apt/mirrors/debian.list Mirrorlist [38 B]
Get:2 file:/etc/apt/mirrors/debian-security.list Mirrorlist [47 B]
Get:3 https://cdn-aws.deb.debian.org/debian trixie InRelease [140 kB]
Get:4 https://cdn-aws.deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Get:5 https://cdn-aws.deb.debian.org/debian trixie-backports InRelease [54.0 kB]
Get:6 https://cdn-aws.deb.debian.org/debian-security trixie-security InRelease [
43.4 kB]
Get:7 https://cdn-aws.deb.debian.org/debian trixie/main Sources [10.5 MB]
Get:8 https://cdn-aws.deb.debian.org/debian trixie/main amd64 Packages [9671 kB]
Get:9 https://cdn-aws.deb.debian.org/debian trixie/main Translation-en [6484 kB]
Get:10 https://cdn-aws.deb.debian.org/debian trixie-updates/main Sources [2788 B
]
Get:11 https://cdn-aws.deb.debian.org/debian trixie-updates/main amd64 Packages
[5412 B]
Get:12 https://cdn-aws.deb.debian.org/debian trixie-updates/main Translation-en
[4096 B]
Get:13 https://cdn-aws.deb.debian.org/debian trixie-backports/main Sources [171
kB]
Get:14 https://cdn-aws.deb.debian.org/debian trixie-backports/main amd64 Package
s [187 kB]
Get:15 https://cdn-aws.deb.debian.org/debian trixie-backports/main Translation-e
n [144 kB]
Get:16 https://cdn-aws.deb.debian.org/debian-security trixie-security/main Sourc
es [123 kB]
Get:17 https://cdn-aws.deb.debian.org/debian-security trixie-security/main amd64
 Packages [114 kB]
Get:18 https://cdn-aws.deb.debian.org/debian-security trixie-security/main Trans
lation-en [72.8 kB]
Fetched 27.8 MB in 2s (11.1 MB/s)
1 package can be upgraded. Run 'apt list --upgradable' to see it.
admin@ip-172-31-7-48:~$
```

admin@ip-172-31-7-48: ~

```
Selecting previously unselected package apache2-data.
Preparing to unpack .../11-apache2-data_2.4.66-1~deb13u2_all.deb ...
Unpacking apache2-data (2.4.66-1~deb13u2) ...
Selecting previously unselected package apache2-utils.
Preparing to unpack .../12-apache2-utils_2.4.66-1~deb13u2_amd64.deb ...
Unpacking apache2-utils (2.4.66-1~deb13u2) ...
Selecting previously unselected package apache2.
Preparing to unpack .../13-apache2_2.4.66-1~deb13u2_amd64.deb ...
Unpacking apache2 (2.4.66-1~deb13u2) ...
Selecting previously unselected package ssl-cert.
Preparing to unpack .../14-ssl-cert_1.1.3_all.deb ...
Unpacking ssl-cert (1.1.3) ...
Setting up libgdbm-compat4t64:amd64 (1.24-2) ...
Setting up libjansson4:amd64 (2.14-2+b3) ...
Setting up ssl-cert (1.1.3) ...
Setting up perl-modules-5.40 (5.40.1-6) ...
Setting up libaprlt64:amd64 (1.7.5-1) ...
Setting up liblua5.4-0:amd64 (5.4.7-1+b2) ...
Setting up apache2-data (2.4.66-1~deb13u2) ...
Setting up libaprutillt64:amd64 (1.6.3-3+b1) ...
Setting up libaprutil1-ldap:amd64 (1.6.3-3+b1) ...
Setting up libaprutil1-dbd-sqlite3:amd64 (1.6.3-3+b1) ...
Setting up libperl5.40:amd64 (5.40.1-6) ...
Setting up perl (5.40.1-6) ...
Setting up apache2-utils (2.4.66-1~deb13u2) ...
Setting up apache2-bin (2.4.66-1~deb13u2) ...
Setting up apache2 (2.4.66-1~deb13u2) ...
Enabling module mpm_event.
Enabling module authz_core.
Enabling module authz_host.
Enabling module authn_core.
Enabling module auth_basic.
Enabling module access_compat.
Enabling module authn_file.
Enabling module authz_user.
Enabling module alias.
Enabling module dir.
Enabling module autoindex.
Enabling module env.
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.

Progress: [ 97%] [                                                  ]
```

Ni varno 3.70.198.144

TrustMeBro

Tega spletnega mesta ni mogoče doseči

Spletno mesto **3.70.198.144** se ni odzvalo v ustreznem času.

Poskusite:

- preveriti povezavo
- preveriti strežnik proxy in požarni zid
- Izvajanje orodja Omrežna diagnostika Windows

ERR_CONNECTION_TIMED_OUT

Znova naloži

Podrobnosti

aws Search [Alt+S] Europe (Frankfurt) enejvolmut (9392-4440-4351) enejvolmut

EC2 > Security Groups > sg-0dc5fdcffddfc0d83 - launch-wizard-2 > Edit inbound rules

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info		Description - optional Info	
sgr-07a9f406ffa6b5d59	SSH	TCP	22	Custom	0.0.0.0/0		Delete
–	HTTP	TCP	80	Anywhe...	0.0.0.0/0		Delete

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel Preview changes Save rules

CloudShell Feedback Console Mobile App

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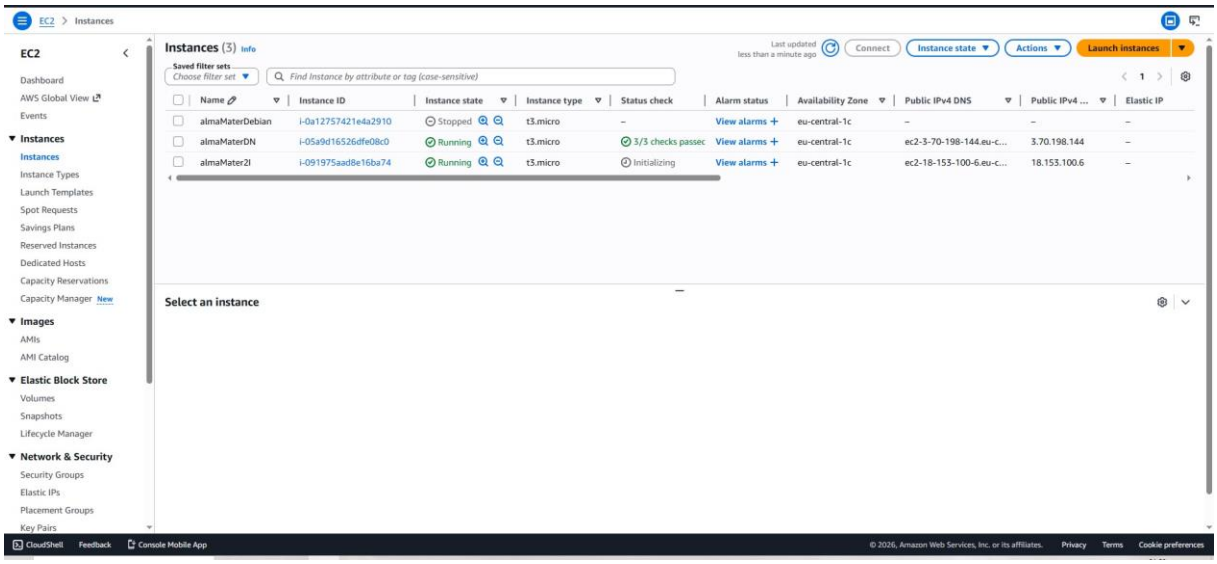
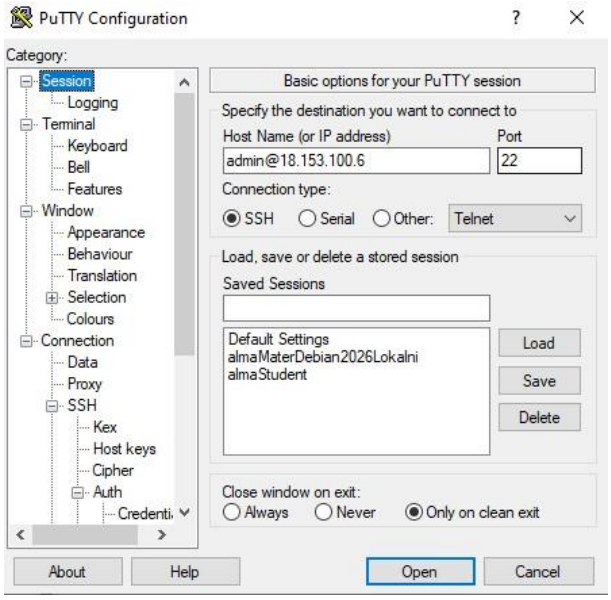
Ni varno 3.70.198.144
TrustMeBro
Apache2 Debian Default Page
debian
It works!
This is the default welcome page used to test the correct operation of the Apache2 server after installation on Debian systems. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should replace this file (located at /var/www/html/index.html) before continuing to operate your HTTP server.
If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.
Configuration Overview
Debian's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Debian tools. The configuration system is fully documented in /usr/share/doc/apache2/README.Debian.gz. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the manual if the apache2-doc package was installed on this server.
The configuration layout for an Apache2 web server installation on Debian systems is as follows:
/etc/apache2/
|-- apache2.conf
| `-- ports.conf
|-- mods-enabled
| |-- *.load
| `-- *.conf
|-- conf-enabled
| `-- *.conf
|-- sites-enabled
| `-- *.conf
apache2.conf is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.
ports.conf is always included from the main configuration file. It is used to determine the listening ports for incoming connections, and this file can be customized anytime.
Configuration files in the mods-enabled/, conf-enabled/ and sites-enabled/ directories contain particular configuration snippets which manage modules, global configuration fragments, or virtual host configurations, respectively.
They are activated by symlinking available configuration files from their respective *-available/ counterparts. These should be managed by using our helpers a2enmod, a2dismod, a2ensite, a2dissite, and a2enconf, a2disconf . See their respective man pages for detailed information.
The binary is called apache2. Due to the use of environment variables, in the default


EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Instances (3) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Find Instance by attribute or tag (case-sensitive)
Name | Instance ID | Instance state | Instance type | Status check | Alarm status | Availability Zone | Public IPv4 DNS | Public IPv4 ... | Elastic IP
almaMaterDebian | i-0a12757421e4a2910 | Stopped | t3.micro | – | View alarms + | eu-central-1c | – | – | –
almaMaterDN | i-05a9d16526dfe08c0 | Running | t3.micro | 3/3 checks passed | View alarms + | eu-central-1c | ec2-3-70-198-144.eu-c... | 3.70.198.144 | –
almaMater2I | i-091975aad8e16ba74 | Running | t3.micro | Initializing | View alarms + | eu-central-1c | ec2-18-153-100-6.eu-c... | 18.153.100.6 | –
Select an instance
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


PuTTY Configuration
Category:
Session
Logging
Terminal
Keyboard
Bell
Features
Window
Appearance
Behaviour
Translation
Selection
Colours
Connection
Data
Proxy
SSH
Kex
Host keys
Cipher
Auth
Credenti
Basic options for your PuTTY session
Specify the destination you want to connect to
Host Name (or IP address)
Port
admin@18.153.100.6
22
Connection type:
SSH
Serial
Other:
Telnet
Load, save or delete a stored session
Saved Sessions
Default Settings
almaMaterDebian2026Lokalni
almaStudent
Load
Save
Delete
Close window on exit:
Always
Never
Only on clean exit
About
Help
Open
Cancel

admin@ip-172-31-2-203: ~

```
 2.4.66-1~deb13u2 [1409 kB]
Get:13 https://cdn-aws.deb.debian.org/debian trixie/main amd64 apache2-data all
2.4.66-1~deb13u2 [160 kB]
Get:14 https://cdn-aws.deb.debian.org/debian trixie/main amd64 apache2-utils amd
64 2.4.66-1~deb13u2 [216 kB]
Get:15 https://cdn-aws.deb.debian.org/debian trixie/main amd64 apache2 amd64 2.4
.66-1~deb13u2 [225 kB]
Get:16 https://cdn-aws.deb.debian.org/debian trixie/main amd64 ssl-cert all 1.1.
3 [16.8 kB]
Fetched 10.1 MB in 0s (72.7 MB/s)
Preconfiguring packages ...
Selecting previously unselected package perl-modules-5.40.
(Reading database ... 30202 files and directories currently installed.)
Preparing to unpack .../00-perl-modules-5.40_5.40.1-6_all.deb ...
Unpacking perl-modules-5.40 (5.40.1-6) ...
Selecting previously unselected package libgdbm-compat4t64:amd64.
Preparing to unpack .../01-libgdbm-compat4t64_1.24-2_amd64.deb ...
Unpacking libgdbm-compat4t64:amd64 (1.24-2) ...
Selecting previously unselected package libperl5.40:amd64.
Preparing to unpack .../02-libperl5.40_5.40.1-6_amd64.deb ...
Unpacking libperl5.40:amd64 (5.40.1-6) ...
Selecting previously unselected package perl.

Progress: [ 10%] [                                                          ]
```

Instance summary for i-091975aad8e16ba74 (almaMater2I) Info

Updated less than a minute ago

Connect | Instance state | Actions

Instance ID i-091975aad8e16ba74	**Public IPv4 address** 18.153.100.6 | open address	**Private IPv4 addresses** 172.31.2.203
IPv6 address –	**Instance state** Running	**Public DNS** ec2-18-153-100-6.eu-central-1.compute.amazonaws.com | open address
Hostname type IP name: ip-172-31-2-203.eu-central-1.compute.internal	**Private IP DNS name (IPv4 only)** ip-172-31-2-203.eu-central-1.compute.internal	
Answer private resource DNS name IPv4 (A)	**Instance type** t3.micro	**Elastic IP addresses** –
Auto-assigned IP address 18.153.100.6 [Public IP]	**VPC ID** vpc-098eca89dceddb091	**AWS Compute Optimizer finding** Opt-in to AWS Compute Optimizer for recommendations. | Learn more
IAM role –	**Subnet ID** subnet-01715b451fa63b8af	**Auto Scaling Group name** –
IMDSv2 Required	**Instance ARN** arn:aws:ec2:eu-central-1:939244404351:instance/i-091975aad8e16ba74	**Managed** false
Operator –		

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

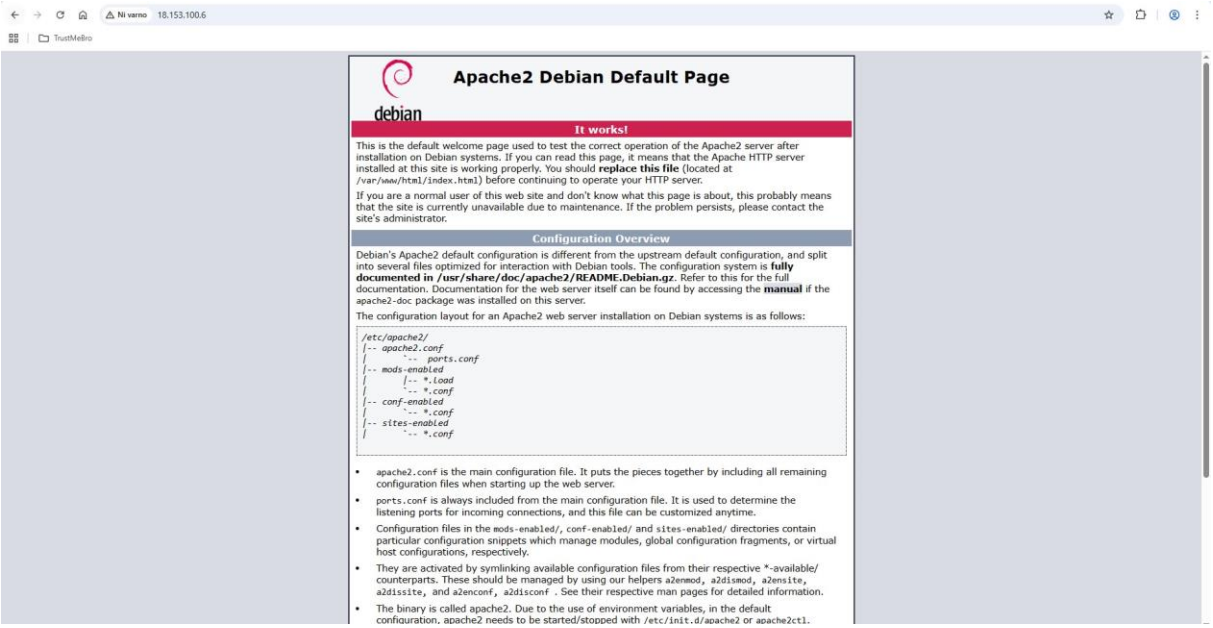
Security details

IAM role –	**Owner ID** 939244404351	**Launch time** Tue Mar 24 2026 21:59:12 GMT+0100 (Srednjeevropski standardni čas)
Security groups sg-00c35e9debdc7694b (launch-wizard-3)		

Inbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Source	Security groups	Description
–	sgr-0a2508ee71174e0c6	22	TCP	0.0.0.0/0	launch-wizard-3	–
–	sgr-0ad01598e886bd077	80	TCP	0.0.0.0/0	launch-wizard-3	–



Windows PowerShell

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\Asus> cd "I:\Alma Mater\RAČUNALNIŠTVO V OBLAKU IN MOBILNE REŠITVE\NALOGE\Predavanje1\splet"
PS I:\Alma Mater\RAČUNALNIŠTVO V OBLAKU IN MOBILNE REŠITVE\NALOGE\Predavanje1\splet> ls


    Directory: I:\Alma Mater\RAČUNALNIŠTVO V OBLAKU IN MOBILNE REŠITVE\NALOGE\Predavanje1\splet


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----       22. 03. 2026     18:16            290 index.html
-a----       15. 07. 2020     14:39        5013504 slika.jpg


PS I:\Alma Mater\RAČUNALNIŠTVO V OBLAKU IN MOBILNE REŠITVE\NALOGE\Predavanje1\splet> scp -o "StrictHostKeyChecking=no" -i "I:\Alma
 Mater\RAČUNALNIŠTVO V OBLAKU IN MOBILNE REŠITVE\NALOGE\almaKljuci\privatniKljucAWS.pem" index.html admin@18.153.100.6:/home/admin
/
Warning: Permanently added '18.153.100.6' (ED25519) to the list of known hosts.

@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@         WARNING: UNPROTECTED PRIVATE KEY FILE!          @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
Permissions for 'I:/Alma Mater/RA\304\214UNALNI\305\240TVO V OBLAKU IN MOBILNE RE\305\240ITVE/NALOGE/almaKljuci/privatniKljucAWS.p
em' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
Load key "I:/Alma Mater/RA\304\214UNALNI\305\240TVO V OBLAKU IN MOBILNE RE\305\240ITVE/NALOGE/almaKljuci/privatniKljucAWS.pem": ba
d permissions
admin@18.153.100.6: Permission denied (publickey).
C:\Windows\System32\OpenSSH\scp.exe: Connection closed                          icacls "I:\Alma Mater\RAČUNALNIŠTVO V OBLAKU
IN MOBILNE REŠITVE\NALOGE\almaKljuci\privatniKljucAWS.pem" /inheritance:r /grant:r "$($env:USERNAME):R"
processed file: I:\Alma Mater\RAČUNALNIŠTVO V OBLAKU IN MOBILNE REŠITVE\NALOGE\almaKljuci\privatniKljucAWS.pem
Successfully processed 1 files; Failed processing 0 files
PS I:\Alma Mater\RAČUNALNIŠTVO V OBLAKU IN MOBILNE REŠITVE\NALOGE\Predavanje1\splet> scp -o "StrictHostKeyChecking=no" -i "I:\Alma
 Mater\RAČUNALNIŠTVO V OBLAKU IN MOBILNE REŠITVE\NALOGE\almaKljuci\privatniKljucAWS.pem" index.html admin@18.153.100.6:/home/admin
/
index.html                                                            100%  290     8.9KB/s   00:00
PS I:\Alma Mater\RAČUNALNIŠTVO V OBLAKU IN MOBILNE REŠITVE\NALOGE\Predavanje1\splet> scp -o "StrictHostKeyChecking=no" -i "I:\Alma
 Mater\RAČUNALNIŠTVO V OBLAKU IN MOBILNE REŠITVE\NALOGE\almaKljuci\privatniKljucAWS.pem" slika.jpg admin@18.153.100.6:/home/admin/

slika.jpg                                                             100% 4896KB  10.7MB/s   00:00
PS I:\Alma Mater\RAČUNALNIŠTVO V OBLAKU IN MOBILNE REŠITVE\NALOGE\Predavanje1\splet>
```

admin@ip-172-31-2-203: ~

```
Enabling module autoindex.
Enabling module env.
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink '/etc/systemd/system/multi-user.target.wants/apache2.service' →
'/usr/lib/systemd/system/apache2.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/apache-htcacheclean
.service' → '/usr/lib/systemd/system/apache-htcacheclean.service'.
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for libc-bin (2.41-12+deb13u2) ...
admin@ip-172-31-2-203:~$ sudo mv /home/admin/index.html /var/www/html/index.html
admin@ip-172-31-2-203:~$ sudo mv /home/admin/slika.jpg /var/www/html/slika.jpg
admin@ip-172-31-2-203:~$
```

Ni varno 18.153.100.6

TrustMeBro

Spletna stran za predmet OblaÄne storitve

Avtor: Enej Volmut

ODGOVORI NA VPRAŠANJA:

1. Kakšna je razlika med NAT in Bridged omrežjem?

Pri NAT omrežju si virtualni stroj deli IP naslov gostiteljskega računalnika in nima lastne neposredne povezave z zunanjim omrežjem. Pri Bridged omrežju pa se virtualni stroj obnaša kot samostojna naprava v lokalnem omrežju, saj dobi svoj lasten IP naslov neposredno od usmerjevalnika (modema).

2. Zakaj v NAT načinu dostop pogosto ne deluje?

Dostop ne deluje zato, ker virtualni stroj nima javno vidnega naslova v lokalni mreži. Zunanji računalniki ne morejo vzpostaviti neposredne povezave z njim, ker je skrit za IP naslovom gostitelja, kateri informacij brez posebnih nastavitev ne zna preusmeriti do virtualnega stroja.

3. Kakšna je razlika med javnim in zasebnim IP naslovom?

Javni IP naslov je unikaten naslov, ki se uporablja za komunikacijo po celem internetu in omogoča dostop do naprave od kjerkoli. Zasebni IP naslov pa se uporablja samo za komunikacijo znotraj lokalnega omrežja in zunaj tega omrežja ni dostopen.

4. Kakšna je vloga SSH?

SSH je varen omrežni protokol, ki omogoča oddaljeno upravljanje računalnika prek ukazne vrstice. Njegova glavna vloga je, da šifrira vso komunikacijo med mojim računalnikom in strežnikom, s čimer preprečuje krajo podatkov med prenosom.

5. Kakšna je vloga Apache strežnika?

Apache je programska oprema, ki deluje kot spletni strežnik. Njegova naloga je, da sprejema zahteve brskalnikov in jim pošlje zahtevane datoteke (npr. HTML strani in slike), da se te lahko prikažejo uporabniku.

6. Kaj omogoča Security Group v AWS?

Security Group v AWS deluje kot omrežni požarni zid za virtualni strežnik. Omogoča določanje pravil, kateri promet (prek katerih vrat in protokolov) sme vstopiti v strežnik in katerega lahko zapusti, s čimer pa omejimo dostop le pooblaščenim uporabnikom in storitvam.